

1. Write a C++ program to Design a **class called STRING** with **data member is char ***

- a) Design proper **constructor** and **destructor** for allocating & de-allocate memory dynamically
- b) Design **set_data() member function** to scan string
- c) Design **get_data() member function** to print string
- c) Design **string_len() member function** to find string length
- d) Design **math_digit() member function** to calculate the sum of even numbers in string and product of odd numbers in string
- e) Design **Palindrome_func() friend function** to check given string palindrome or not?

ex: "vector123india56"

o/p: string length: 9

sum: 8

product: 15

Given string is not palindrome

2. Write a C++ program to Design a **class called STRING** with **data member is char ***

- a) Design proper **constructor** and **destructor** for allocating & de-allocate memory dynamically
- b) Design **set_data() member function** to scan string
- c) Design **get_data() member function** to print string
- c) Design **count_word() member function** to find word count in string
- d) Design **Reverse_str() friend function** to reverse each word from a given string(**Just not Reverse Print**)

Ex: "vector India in Bangalore"

o/p: word count: 4

reverse word: Bangalore in India Vector

1. Write a C++ Program to define a class called **STUDENT** with **data members** as rollno , Name and Marks. Define appropriate **Member functions** 1) Set_data(), 2) get_data() for Reading and displaying the data and also define a member function called **Sort_data()** to Sort 5 students records according to the rollno , Name and Marks.

Note: Data member must be declared in Private Section.

2. Write a C++ program to declare 3 classes .Add Two class data member array values into third class, using friend function .

Note: Data member must be declared in Private Section.

1. Write a C++ program To design a class Called STRING with the Data member as char* and proper member functions for (1).constructor (2).destructor (3) set_data(), (4) get_data() (5) string Palindrome() (6) Design StrStr() Palindrome example:

input :- Malayalam

output:- Malayalam

StrStr example

input :- ROHIT HIT ROHIT

output:- HIT (PRESENT COUNT=3)

Note: Data member must be private section.

2. Write a C++ program To design a class Called STRING, Data member as char* and proper member functions for (1).constructor (2).destructor (3) set_data() (4) get_data() (5) Design strrev() (6) Delete duplicate character in a given string

Delete example:

input:- VECTEEOVVORR

output:- VECTOR

Strrev Example

input:- exam

output:- maxe

Note: Data member must be private section

1.WAP to Perform Multiplication on two matrices and store the result in third matrix .Allocate the memory dynamically for all the matrices. If matrix multiplication not possible then display usage message on screen.

Note: scan the rows and columns for first and second matrices

1.Design a class called an ARRAY with data member as int *p .

Use appropriate member functions **set_data()** and **get_data()** , **sum_of_prime()** as a member function; **Armstrong_num()** as a **friend function**;

Note : data member must be in private section.

2a Write a program to demonstrate the **Shallow copy** .

2b. Write a program to demonstrate the **Deep copy** .

Note: Data Members must be in private section.